Cambridge O Level

Cambridge Assessment International Education
Cambridge Ordinary Level

PHYSICS **5054/42**

Paper 4 Alternative to Practical **October/November 2019**

MARK SCHEME

Maximum Mark: 30

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the October/November 2019 series for most Cambridge IGCSE™, Cambridge International A and AS Level components and some Cambridge O Level components.

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Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

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GENERIC MARKING PRINCIPLE 5: Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).
GENERIC MARKING PRINCIPLE 6: Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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<table>
<tr><th>Question</th><th>Answer</th><th>Marks</th></tr>
<tr><td>1(a)(i)</td><td>66 ± 1 (mm)
<table><tr><td>L / N</td><td>L / mm</td></tr><tr><td>1.0</td><td>66</td></tr><tr><td>2.0</td><td>100</td></tr><tr><td>3.0</td><td>130</td></tr><tr><td>4.0</td><td>163</td></tr><tr><td>5.0</td><td>198</td></tr></table></td><td>B1</td></tr>
<tr><td>1(a)(ii)</td><td>rule drawn (vertical, by eye) close (10 mm or less from either side of the spring) to the spring and longer than it</td><td>B1</td></tr>
<tr><td>1(a)(iii)</td><td>middle of the X level with the ‘bottom’ of the spring</td><td>B1</td></tr>
<tr><td>1(b)(i)</td><td>list readings in order of increasing / decreasing load</td><td>B1</td></tr>
<tr><td rowspan="4">1(b)(ii)</td><td>axes labelled quantity and unit and axes correct way round</td><td>B1</td></tr>
<tr><td>scales linear, not awkward and start from (0,0)</td><td>B1</td></tr>
<tr><td>points plotted accurately, to the nearest $\frac{1}{2}$ square</td><td>B1</td></tr>
<tr><td>best-fit thin straight line drawn</td><td>B1</td></tr>
<tr><td>1(b)(iii)</td><td>no and line does not pass through the origin</td><td>B1</td></tr>
<tr><td>1(b)(iv)</td><td>graph extrapolated to cut the x-axis and $l_0 = 34 \pm 2$ (mm)</td><td>B1</td></tr>
<tr><td rowspan="2">1(c)(i)</td><td>candidate’s value at 3.6 read from the graph</td><td>B1</td></tr>
<tr><td>above value – candidate’s (b)(iv)</td><td>B1</td></tr>
</table>

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Question	Answer	Marks
1(c)(ii)	straight line through the origin	**B1**
1(d)	same initial value of l	**B1**
	straight line and steeper slope	**B1**

Question	Answer	Marks
2(a)	13.6 (g)	**B1**
2(b)(i)	79(.0) (cm^3)	**B1**
2(b)(ii)	14(.0) (cm^3)	**B1**
2(b)(iii)	not reading perpendicular to the scale / at eye level not reading to the bottom of the meniscus / measuring cylinder only reads to (1 or) 2 cm^3	**B1**
2(c)	0.971	**B1**
	g / cm^3	**B1**

Question	Answer	Marks
3(a)(i)	0.21 (V)	**B1**
3(a)(ii)	0.84 (Ω)	**B1**
3(b)	as l increases R increases / they are proportional	**C1**
	l and R are directly proportional / l / R is constant / doubling l doubles R, etc.	**A1**
3(c)	switch off between readings / use smaller currents / voltages / use a fan	**B1**

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Question	Answer	Marks
4(a)(i)	normal at 90° to prism surface at Z	**B1**
4(a)(ii)	ray from Z parallel to WY to hit side XY	**B1**
	ray perpendicular to WY and passing through P_3 and P_4	**B1**
4(b)	reverses the ray / turns the ray through 180° / in the opposite direction	**B1**

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